

EcoSort AI

Smart Waste Segregation Assistant

A sustainability-focused web prototype that helps users identify waste categories and choose more responsible disposal actions.



12

Primary SDG

SDG 12 — Responsible
Consumption

11

Secondary SDG

SDG 11 — Sustainable Cities

1M1B AI for Sustainability Virtual Internship

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Problem background

Why a simple waste-segregation assistant is useful

1

Waste confusion

Users may be unsure whether everyday items belong in wet, recyclable, e-waste or special-handling categories.

2

Mixed disposal

Incorrect segregation can contaminate materials and make downstream sorting and recovery more difficult.

3

Special materials

Batteries, electronics, glass and other items may require safer or separate handling.

Opportunity: Provide quick, understandable guidance at the point where the disposal decision is made.

Problem statement

A clear design question

How might we use AI to help people correctly identify and segregate waste so that waste management becomes more sustainable?



WHO

Students, households, offices
and communities



CHALLENGE

Uncertainty about the right
waste category

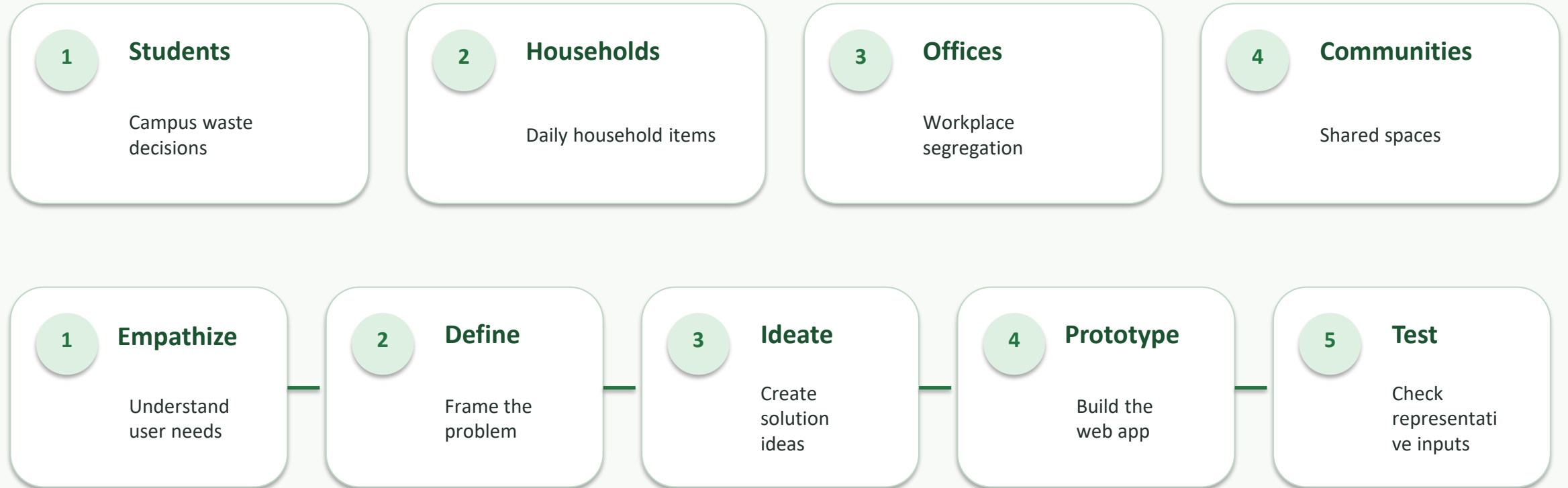


GOAL

Simple guidance for responsible
disposal

Target users & design thinking

Human-centered development



The prototype was designed around a simple user journey: describe an item → receive a category → understand why → act responsibly.

SDG alignment

Primary and secondary sustainability connections

12

SDG 12

RESPONSIBLE CONSUMPTION AND PRODUCTION

- Encourages responsible waste segregation
- Supports recycling awareness
- Gives disposal recommendations
- Highlights special-handling waste

11

SDG 11

SUSTAINABLE CITIES AND COMMUNITIES

- Supports cleaner community practices
- Promotes sustainability awareness
- Can complement local waste-management efforts

Proposed solution

EcoSort AI — a browser-based decision-support prototype

EcoSort AI

Enter a waste item

used plastic water bottle

Analyze

Result

Dry / Recyclable Waste

1

Identify

Classify the described item into a practical waste category.

2

Explain

Give a reason and disposal recommendation.

3

Encourage

Show sustainability tips, history and eco points.

AI workflow & architecture

Input → classification → guidance → action



Waste categories supported by the prototype



Prototype features

Core functionality included in the web application

1

Waste analyzer

Enter an item and receive a category.

2

Recommendations

Reason + disposal guidance.

3

Safety alerts

Warnings for special materials.

4

History

Recent analyses stored locally.

5

Eco points

Simple engagement and progress.

6

Dark mode

Alternative interface appearance.

Prototype testing

Representative examples for functional checking

1

Banana peel

Wet / Organic

2

Plastic bottle

Dry / Recyclable

3

Newspaper

Dry / Recyclable

4

Old mobile phone

E-Waste

5

Used battery

Hazardous / Special

6

Food leftovers

Wet / Organic

7

Cardboard box

Dry / Recyclable

8

Broken glass

Glass / Special

Important: these are prototype test cases, not a measured machine-learning accuracy benchmark.

Responsible AI

Safety and transparency are built into the project



Fairness

Use consistent classification rules across users.



Transparency

Clearly state the prototype's current AI limitations.



Privacy

Minimize data collection; history is stored locally.



Safety

Flag hazardous/special items and encourage verification.



Human oversight

Ambiguous items should be checked by people.



Future validation

A real AI model needs accuracy, bias, privacy and safety testing.

Expected impact & limitations

What can be demonstrated — and what still needs improvement

E

Environmental

Can encourage better segregation and recycling awareness.

S

Social

Can make sustainability guidance easier to understand.

P

Practical

Can provide quick support for common waste decisions.

CURRENT LIMITATIONS

- Current classification uses predefined/curated logic; it is not a trained ML model.
- Ambiguous or mixed-material items may require human verification.
- Local waste-management rules can differ by location.
- Image upload is currently a preview feature; real image classification is future scope.

Future scope & conclusion

Path from a student prototype to a validated sustainability assistant



FUTURE DEVELOPMENT

- Real AI/ML text and image classification
- Multimodal, voice and multilingual support
- Location-specific disposal rules
- Authorized collection-point integration
- Mobile application and analytics



CONCLUSION

EcoSort AI demonstrates a simple digital approach to helping users understand waste categories and disposal choices.

Its next stage would require a validated AI model, real-world testing, local waste rules and continuous safety evaluation.

THANK YOU

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